Rounak Gera

gerarounak890@gmail.com | +91 8968515468 | Khanna, Punjab

linkedin.com/in/rounak-gera-229a46287 | github.com/rounak890 | rounak-gera.carrd.co

Education

Thapar Institute of Engineering & Technology, Masters in Computer Applications 2026 – 2028

Amity University Punjab, Bachelor's in Computer Applications (GPA: 9.33/10.0) 2023 – 2026

L.S.M.S Vidya Mandir, Khanna, Punjab, 10+2 (Percentage: 98.2%) 2022 – 2023

Internships

Summer Intern - Design Innovation Centre (DIC), Panjab University Jun 2026 – Jul 2026

- Improved **cranial reconstruction** performance by **4.5%** through optimized medical image preprocessing, volumetric data pipelines, and a lightweight 3D deep learning architecture.
- Designed a **deployment-ready lightweight 3D U-Net** with optimized inference and low computational overhead while maintaining high reconstruction accuracy; the project received the **Best Project Award**.

Freelance Deep Learning Specialist Sep 2023 – February 2024

- Optimized deep learning models for edge deployment using pruning, quantization and ONNX Runtime, reducing inference time by up to **40%**.
- Applied model compression techniques to reduce model size by **20%** while targeting efficient resource-constrained inference.

Projects & Research Work

MiniRAG: MSA Contract Question Answering System - GitHub Repo May 2026 - June 2026

- Built an end-to-end **RAG pipeline** for contract QA using SentenceTransformers embeddings, FAISS retrieval, and LLM inference.
- Implemented recursive chunking, query expansion and lightweight reranking to improve retrieval under contract terminology mismatch; evaluated QA using an **LLM-as-a-Judge** framework.
- *Tools Used: Python, FAISS, Ollama, PyPdf, Sentence Transformers*

Smart Assistant for Chartered Accountants (CA Buddy) - GitHub Repo June 2025 - Aug 2025

- Built an **AI-powered assistant** for Chartered Accountants, combining document intelligence, tax computation and compliance automation into a unified workflow.
- Built **retrieval** and **document-processing** workflows using **FAISS**, **Tesseract OCR** and **Gemini API** to automate accounting tasks.
- *Tools Used: Python, Django, React, Gemini API, Tesseract OCR, Pandas and FAISS.*

MADETFE : Mask aware dual expert with transformers and FFC based high-fidelity image inpainting(Applied Intelligence) *(Second Author)*

- Proposed a **sequential LaMa–MAT framework** combining FFCs and Mask-aware Transformers for high-fidelity image inpainting.
- Outperformed current SOTA image inpainting models in all metrics by almost **15%** through ensemble learning and transfer learning.

Multimodal Pneumonia Detection from Chest X-Rays and Patient Metadata Using Multi-Stage Pretraining and Fine-Tuning (ICT4SD' 2026) *(First Author)*

- Developed a **multimodal pneumonia detection framework** integrating **chest X-ray images** and **patient metadata**.
- Proposed a **multi-stage pretraining and fine-tuning pipeline** with **Global Weighted Average Pooling (GWAP)** for improved feature localization. Along **Grad-CAM** and **SHAP**-based explainability analysis.

Technical Skills & Relevant Coursework

- **Programming & ML:** Python, C++, Java, PyTorch, Scikit-learn, NumPy, Pandas
- **Generative AI & RAG:** Hugging Face, FAISS, SentenceTransformers, Gemini API, Ollama
- **Computer Vision:** OpenCV, Grad-CAM, SHAP
- **Frameworks:** Django, FastAPI, React
- **Certifications:** Oracle Cloud Infrastructure Generative AI Professional, Oracle Cloud Infrastructure AI Foundations Associate, AWS Educate Machine Learning Foundation

Awards & Achievements

- **Secured 2nd Runner-Up position at RE-IGNITE 2.0 Ideathon 2026 organized by AUP, Mohali** – Feb 2026
- **Achieved 466 rank in TCS CodeVita Season 12 organized by TCS** – January 2025
- **Secured 18th Place at Build Fast with AI Hackathon organized by Build Fast with AI** – 2024
- **Recieved FAP National Award for Academic Excellence by the Federation of Academic Professionals** – 2023
- **Student Placement Representative (SPR)**, Career Guidance & Placement Cell, Amity University Punjab (2026) – Actively contributed to student management, placement coordination, industry outreach, and engagement activities.